Local-KLE (MSM) pCN samplers, round 2: speed versus posterior fidelity

4 November 2026

1 Summary of findings

This report covers the 15 runs of Table 1: the second-round experiments and the re-run first-pass experiments (all 64 chains, 10^5 iterations, $M = 10^4$ adaptation iterations, state saved every 50 iterations except experiments 4 and 5, saved every 100). The setup, rank test and prior test are in the first report (`msm_local_kle_report.pdf`) and are not repeated.

- **Only experiment 3b reproduces the global posterior.** Its mean posterior standard deviation of $\log \kappa$ is 0.576 against 0.570 for global pCN, and its error of the posterior-mean field is 0.409 against 0.416. Every other local-KLE sampler is narrower: by 10–17% (ratios 0.83–0.90) for all except the mixtures 4 and 4.8, which collapse (ratios 0.38 and 0.44). Their mean fields are also further from the truth (0.42–0.50, and 0.56 for the mixtures).
- **Speed and fidelity trade off, as you expected.** 3b reaches MPSRF < 1.2 after 51k iterations against 56k for global pCN (about 9% faster, a single replicate, so within plausible noise). The under-dispersed samplers (excluding the mixtures) reach it after 28–45k. The fastest honest alternative, alternating with no offset (6d), needs 29k, about 48% fewer iterations than global pCN, but it samples a distribution about 14% too narrow.
- **Experiment 4 is not rescued by a larger local dimension, and now we know why.** With 8 local modes the primal acceptance rises from 0.16% to 25%, but the posterior spread is still only 0.44 of the global one and the median relative error is 0.030 against the 0.0204 plateau. The round trip $B_g \Pi_g$ it applies contracts (mean eigenvalue 0.76 primal, 0.93 dual) and the chains sit at $|\theta|^2 \approx 7$–17 against about 55 for global pCN, each at its own persistent level (Section 3.2). Alternating through the global state with Π_g should be dropped.
- **The innovation form of the mixture (4b) and all alternating schemes (6a–6d) are well behaved**: acceptance $\approx 25\%$ in both grids, plateau relative error 0.0203–0.0204, MPSRF below 1.1 in the 58 range coordinates. They share the under-dispersion (ratio 0.86–0.89) of the raw primal sampler and are not prior preserving.
- **The number of inner iterations L and the frozen-offset move matter little.** $L = 1, 10, 100$ reach MPSRF < 1.2 at 37k, 36k, 42k; dropping the offset (6d) is fastest (29k) but also gives a slightly narrower posterior (0.86) than the offset versions (0.88–0.89).

2 Experiments and how MPSRF is computed

1	global pCN on $\theta \in \mathbb{R}^{64}$
2, 2.6, 2.8	primal raw, $\theta = Bc$; 4, 6, 8 local modes per 4×4 block ($p = 64, 96, 128$)
3a, 3b	whitened pseudo-inverse; whitened plus null-space term $V_0\zeta$ (exact prior)
4, 4.8	mixture primal/dual through the global state with projection Π_g (4 and 8 local modes)
4b	mixture, innovation form $\theta' = \sqrt{1-s^2}\,\theta + s\,B_g\xi$
5, 5b	$5 \cdot 10^4$ primal iterations then $5 \cdot 10^4$ dual; switch by Π_{dual} (5) or the exact lift L_d (5b)
6a–6c	alternating primal/dual blocks of $L = 1, 10, 100$ iterations, frozen-offset move $\theta' = \theta + B_g(c' - c_0)$, $c_0 = L_g\theta$
6d	as 6a but without the offset ($\theta' = B_g c'$ in the primal block)

MPSRF and PSRF (group convention, `brho/mpsrf.py`). The window starts at iteration 1000 (of the phase, for phase 2 of experiment 5), so the adaptation period is *not* discarded, and it grows to the end of the run: at each curve point it holds the saved samples from the start to the current iteration, every 2nd sample, with a curve point every 15 saved samples. W is the average within-chain covariance (divisor $n-1$), B/n the covariance of the chain means (divisor $m-1$), MPSRF $= \sqrt{(n-1)/n + (m+1)/m\,\lambda_{\max}(W^{-1}B/n)}$, and $\text{PSRF}_i = \sqrt{V_{ii}/W_{ii}}$ with the maximum over components reported. Not split-chain, no extra burn-in. "Iterations to 1.2" is the first curve point (ignoring the first 10) below 1.2, linearly interpolated. For comparability all runs are measured on the same 58 coordinates $V_+^\top\theta$ of the rank-58 range ("common 58"); for experiment 2 and 3a, whose θ lies in that range, these are the natural coordinates, and for experiment 2.6, 2.8 and 4.8 the 64-dimensional θ is used (the common-58 entry coincides with it). For phase 2 of experiments 5 and 5b the count is in iterations after the switch.

3 Results

run	acc. (primal/dual)	median relerr	iter. to 1.2 (k)	MPSRF end	max PSRF end	mean sd $\log\kappa$	ratio to 1	rms err. mean
1 global	0.25	0.0205	56.1	1.115	1.052	0.570	1.00	0.416
2 primal raw d4	0.25	0.0204	37.2	1.072	1.033	0.474	0.83	0.496
2.6 primal raw d6	0.26	0.0204	45.2	1.081	1.044	0.491	0.86	0.479
2.8 primal raw d8	0.26	0.0204	37.7	1.072	1.039	0.500	0.88	0.477
3a pinv	0.25	0.0204	45.2	1.083	1.042	0.512	0.90	0.457
3b aug	0.26	0.0205	51.2	1.110	1.048	0.576	1.01	0.409
4 mixture d4	0.00/0.25	0.0332	22.7	1.397	1.084	0.217	0.38	0.556
4.8 mixture d8	0.25/0.25	0.0300	15.9	1.280	1.037	0.250	0.44	0.562
4b mixture innov.	0.25/0.24	0.0203	37.3	1.064	1.031	0.500	0.88	0.431
5 sequence	0.25/0.27	0.0205	33.0	1.131	1.067	0.491	0.86	0.446
5b sequence exact	0.25/0.26	0.0205	27.9	1.125	1.053	0.491	0.86	0.451
6a alt L=1	0.25/0.25	0.0204	37.1	1.085	1.035	0.504	0.88	0.425
6b alt L=10	0.25/0.26	0.0204	35.8	1.077	1.035	0.508	0.89	0.427
6c alt L=100	0.26/0.27	0.0204	41.6	1.072	1.031	0.504	0.88	0.421
6d alt L=1 no offset	0.25/0.23	0.0204	29.2	1.060	1.036	0.490	0.86	0.440

Table 1: All runs. Fields use 10,000 pooled samples after adaptation (5,000 per phase for experiments 5, 5b, pooled here). "ratio to 1" is the mean posterior sd of $\log\kappa$ divided by that of experiment 1; "rms err. mean" is the rms difference between the posterior-mean $\log\kappa$ field and the truth. For experiments 4 and 4.8 the "iter. to 1.2" entry is a false crossing (Section 3.2).

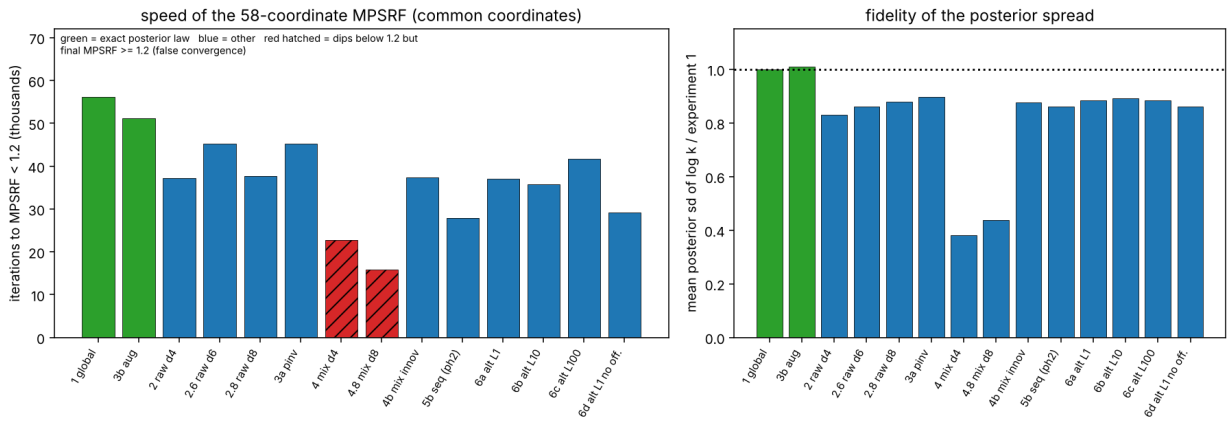


Figure 1: Left: iterations to MPSRF < 1.2 (58 common coordinates). Right: mean posterior sd of $\log \kappa$ relative to global pCN. The two samplers that preserve the posterior (green) are the slowest; everything faster is also narrower.

3.1 Speed comes with variance loss

The relationship in Figure 1 is the central result. The raw primal sampler (2) and the whitened pseudo-inverse sampler (3a) are faster than global pCN, and both have the narrower posterior of the model they actually sample, $N(0, \Sigma_{ms})$ or $N(0, V_+ V_+^\top)$. In the standard-deviation maps (Figure 2) experiment 2 shows an imprint of the 4×4 blocks that is absent in the global and 3b maps, and the maps of the alternating and innovation samplers (6a, 6d, 4b) are close to each other and slightly flatter than the global one. Adding the null-space term (3b) restores the target exactly and takes the speed advantage with it, apart from a modest remainder.

Eight local modes per block (2.8) raises the ratio from 0.83 to 0.88 and keeps the speed of experiment 2 (37.7k against 37.2k); six modes (2.6) is in between in spread but slower (45k). With one replicate per setting I would not read the ordering $37 < 45 < 38$k among 2, 2.6 and 2.8 as meaningful.

3.2 Experiment 4: why a larger local dimension does not fix it

In experiment 4 the primal proposals are almost never accepted (0.16% against 25% for dual) and the median relative error is 0.033, above the posterior plateau. Its MPSRF curve (Figure 5, left) dips to about 1.18 near 25k iterations and then rises to 1.40, so the "22.7k" in Table 1 is a transient crossing, not convergence. With 8 local modes (4.8) both grids accept at 25%, but the sd ratio is only 0.44, the median relative error is 0.030 and the final MPSRF is 1.28. The acceptance problem was therefore a symptom, not the cause. The per-chain traces of θ (Figure 3) and the round-trip map explain what happens instead.

The chains are contracted. For global pCN $|\theta|^2$ fluctuates around about 55 (the pilot runs give 54.95), with every chain overlapping. For experiments 4 and 4.8 it sits at about 7–16 and 10–17 (read from the trace plots, not computed from the chain arrays), i.e. roughly 13–31% of the global value, in every chain. In individual coordinates the spread is about half the global one, and some means move toward zero (for example θ_{30} is near -1.4 for global pCN and near -0.5 in experiment 4).

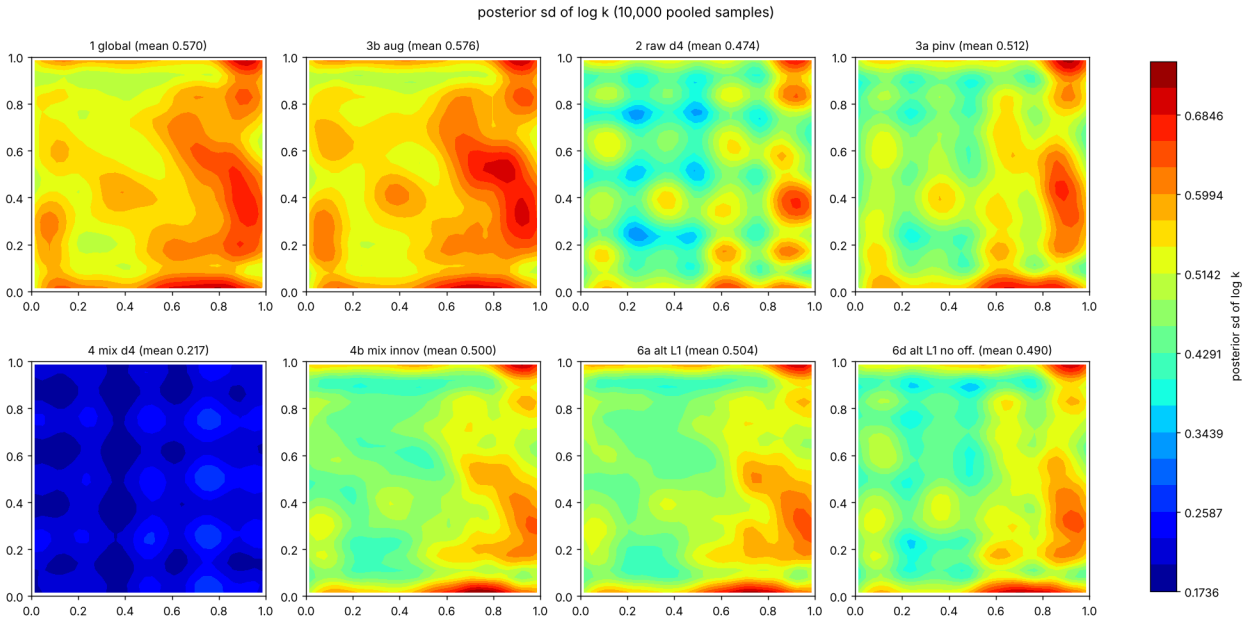


Figure 2: Posterior sd of $\log \kappa$, shared colour scale (jet, contourf), eight representative runs.

The round trip contracts. At each accepted move the mixture applies $\theta \mapsto B_g \Pi_g \theta$ to the part of θ it proposes in. A projection would have eigenvalues 0 or 1 only. The eigenvalues of $B_g \Pi_g$ are strictly between (Table 2, Figure 4): the primal map has mean 0.76 (six zeros, 61% of the eigenvalues below 0.95, smallest non-zero 0.54), the dual map has mean 0.93 and smallest eigenvalue 0.66. The maps are not normal (largest primal singular value 1.21), so they contract in the eigenvalue sense, not in every norm. With acceptance based only on the likelihood ratio nothing restores the shrinkage in directions the data do not constrain.

grid	mean eigenvalue	smallest	fraction < 0.95	largest singular value
primal ($p = 64$)	0.756	≈ 0 (6 directions)	0.61	1.206
dual ($p = 114$)	0.931	0.657	0.38	1.021

Table 2: Spectrum of the round-trip map $B_g \Pi_g$ (64×64) used by experiments 4 and 4.8 (`experiments/msm_roundtrip_spectrum.py`).

The chains do not agree on how far from the origin to sit, though they agree on direction. Ordering the chains by median relative error, the $|\theta|^2$ levels keep that order for the whole 90k post-adaptation iterations (darkest chains at about 15, lightest at about 8–10 in experiment 4; Figure 3). In 4.8 the chains' median relative errors range from 0.027 to 0.033, whereas the global chains share a single level (0.0204) with no ordering. The chain means do not differ in direction: in 4.8 the between-chain sd of the chain means is 0.02–0.1 per coordinate against a within-chain sd of 0.12–0.47, and the largest per-coordinate PSRF is only 1.04, while the MPSRF is 1.28 (experiment 4: 1.07 and 1.40). The MPSRF is high because $\lambda_{\max}$ picks out a combination of coordinates, which is consistent with a chain-level difference in overall size. *Why* each chain settles at its own radius and stays there is not explained by these runs; it is a hypothesis that the contraction strength depends on each chain's history of accepted moves, which I have not tested.

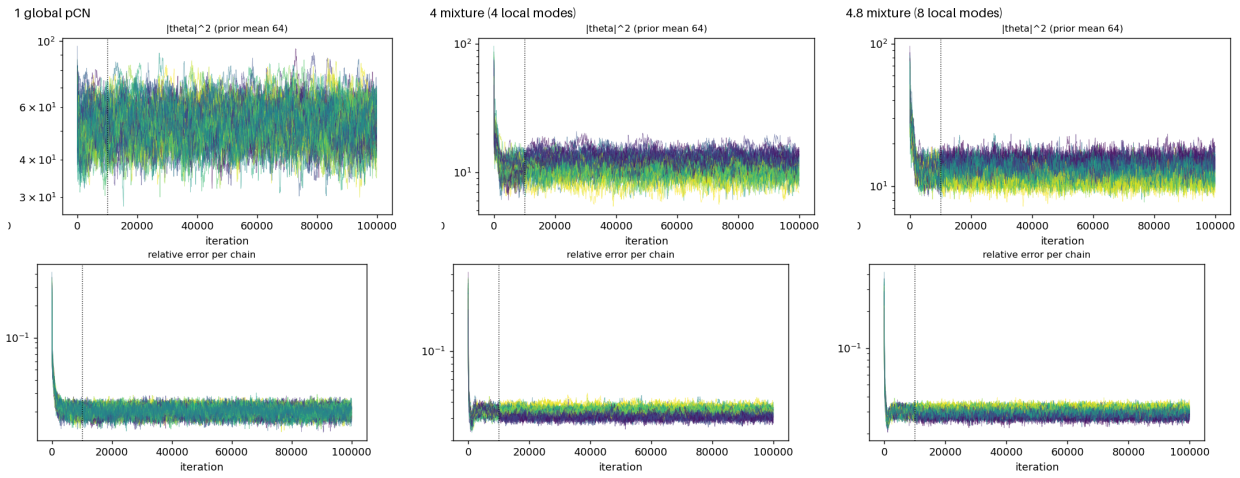


Figure 3: $|\theta|^2$ (top, log scale; prior mean 64) and relative error (bottom) for all 64 chains, coloured by the chain's median relative error (dark = low). Global pCN overlaps with no ordering; both mixtures sit far below it and keep a persistent chain ordering. Panels cropped from `fig_theta_traces.png` of each run.

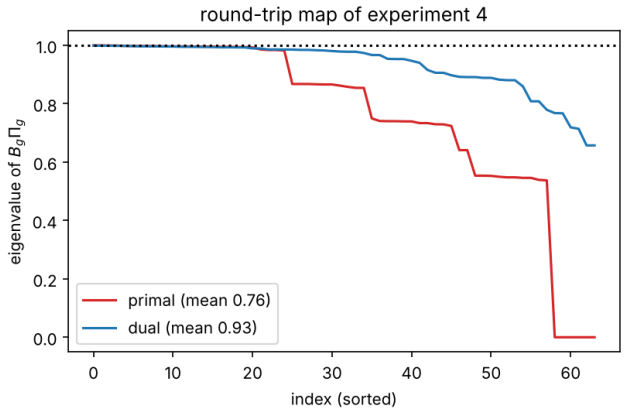


Figure 4: Sorted eigenvalues of $B_g\Pi_g$ for the primal and dual grids.

Conclusion. Experiment 4 as defined (alternating through the global state with the projection Π_g) does not work, at 4 or at 8 local modes. This does not rule out alternating primal/dual updates in general: the samplers that avoid the round trip (4b, 6a–6d) reach the plateau and mix, but sample a posterior about 12–14% too narrow.

3.3 Alternating samplers and the primal→dual sequence

The alternating schemes are not prior preserving: the earlier stationary-covariance analysis gave a pointwise *prior* variance ratio of about 0.77 for them. The measured *posterior* sd ratios here (0.86–0.89, variance ratios 0.74–0.79) are of the same size, although they are different quantities. Block length L and the offset change the speed a little (29–42k) and the spread hardly at all. For the sequence, phase 1 reaches 1.2 at about 37k iterations and phase 2, measured in the window beginning 1000 iterations after the switch, at 33k (5) and 28k (5b, exact lift) iterations after the switch. The difference between 5 and 5b is of a size I would expect from run-to-run noise.

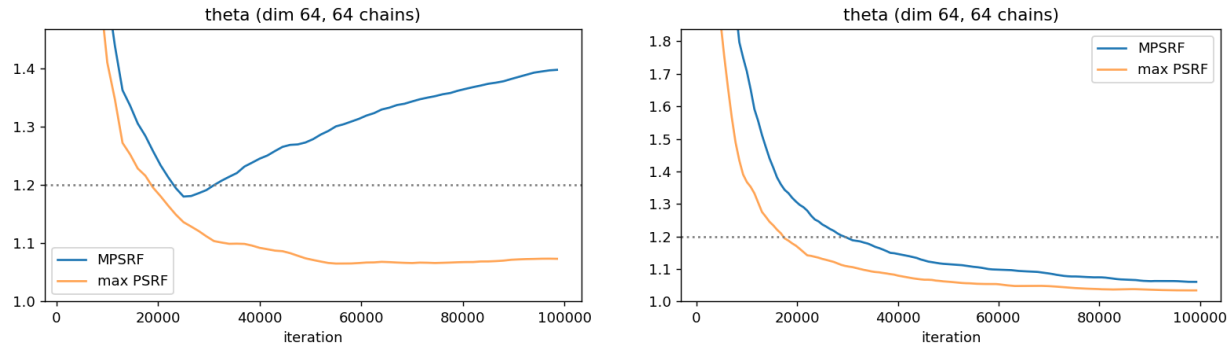


Figure 5: MPSRF and max PSRF of θ (64 dimensions). Left: experiment 4 (dips below 1.2, then rises). Right: experiment 6d (monotone decrease, crosses 1.2 at about 29k).

3.4 Mean-field accuracy

The error of the posterior-mean $\log \kappa$ field relative to the truth is lowest for 3b (0.409) and global pCN (0.416), 0.42–0.46 for the alternating, innovation, 3a and sequence samplers, 0.48–0.50 for the raw primal runs, and 0.56 for the mixtures. Figure 4 shows the corresponding mean maps. The differences among the middle group are small compared with the noise expected from a single data set and seed.

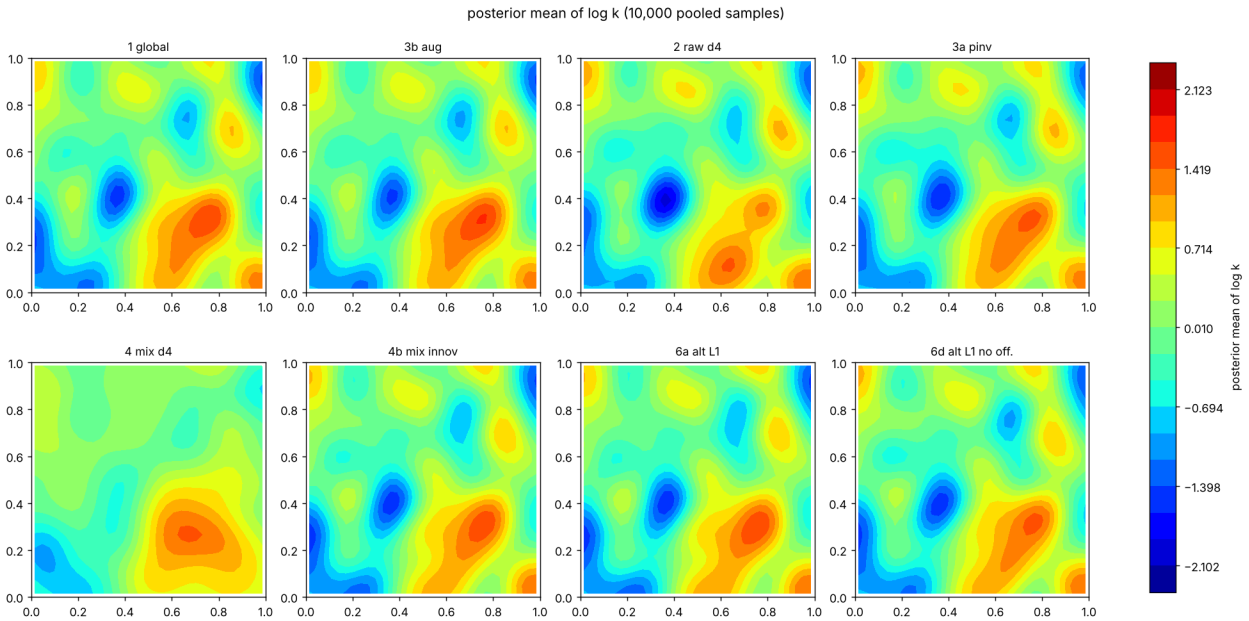


Figure 6: Posterior mean of $\log \kappa$ (shared scale) for the same eight runs.

4 Caveats

- **One replicate, one data set.** Every run uses 64 chains from a single seed set and one noisy data set. Differences of 10–15% in iterations to 1.2 (for example 5 against 5b, or 2 against 2.8) are not distinguishable from noise here; the speed differences of 20–48% for the

non-mixture samplers relative to global pCN and the sd ratios of 0.83–0.90 against 1.01 for 3b are larger, but a second seed set would confirm them.

- **The sd comparison uses correlated samples.** It is a pooled posterior sd of 10^4 saved states; the ratio of 3b to 1 (1.01) is consistent with a Monte Carlo error of order a percent, but one pair of runs cannot establish it.
- **MPSRF in the 114 dual coordinates.** The post-processing also reports MPSRF of the dual local coefficients $\Pi_d\theta$ or $L_d\theta$ (114 coordinates, rank 64, files `post_summary.json` entries `c_dual_proj`), with values up to 1.95. Their within-chain covariance is singular, so these numbers are numerically unreliable and I do not use them. The 64-coordinate primal projections (rank 58) share the issue in milder form. The *state* coefficients of experiments 2, 2.6, 2.8, 3a, 3b and phase 1 of 5 are full rank and are reported in `post_summary.json`.
- **The fidelity measure is the sd of $\log\kappa$ only.** I did not compare higher moments or the posterior of the data fit.

5 Interpretation and next steps

For the rank-58 primal basis, local-KLE pCN is faster than global pCN only to the extent that it fails to sample some of the long-range variance; with the exact prior restored (3b) the gain is small. If the aim is a fast, exact sampler, the results suggest looking for a version of the alternating scheme with a correction step that restores the prior covariance; this has not been tested. If an approximate posterior with bounded variance loss is acceptable, the alternating no-offset sampler (6d) gives the fastest mixing here. For experiment 4 as defined (projection through the global state), the results argue for abandoning it in favour of 4b.

Reproduction. The scripts `make_msm_round2_table.py` and `make_msm_round2_figures.py` (in `experiments/`) read only the committed `summary.json`, `post_summary.json` and `fields.npz` files. Run them with the environment variable `BRHO_ROOT` set to `runs/msm_local_kle_2026-11`.